MCS-213	NETWORK SECURITY & CRYPTOGRAPHY	L T P 3 1 0	4 Credits

Course Objectives:

CO1	To understand basics of Cryptography and Network Security
CO2	To be able to secure a message over insecure channel by various means.
CO3	To learn about how to maintain the Confidentiality, Integrity and Availability of a data.
CO4	To understand various protocols for network security to protect against the threats in the networks.
CO5	To understand various IP Securities

MODULE I

Introduction: OSI Security Architecture - Classical Encryption techniques – Cipher Principles – Data Encryption Standard – Block Cipher Design Principles and Modes of Operation Evaluation criteria for AES – AES Cipher – Triple DES – Placement of Encryption Function – Traffic Confidentiality

MODULE II

Public Key Cryptography: Key Management - Diffie-Hellman key Exchange – Elliptic Curve Architecture and Cryptography - Introduction to Number Theory – Confidentiality using Symmetric Encryption – Public Key Cryptography and RSA. Authentication and Hash Function: Authentication requirements – Authentication functions – Message Authentication Codes – Hash Functions – Security of Hash Functions and MACs – MD5 message Digest algorithm - Secure Hash Algorithm – RIPEMD – HMAC Digital Signatures –Authentication Protocols – Digital Signature Standard

MODULE III

Network Security Authentication Applications: Kerberos – X.509 Authentication Service – Electronic Mail Security – PGP – S/MIME - IP Security – Web Security. System Level Security Intrusion detection – password management – Viruses and related Threats – Virus Counter measures – Firewall Design Principles – Trusted Systems.

References:

1. *Atul Kahate, "Cryptography and Network Security", Tata McGraw-Hill, 2003.*
2. *Bruce Schneier, "Applied Cryptography", John Wiley & Sons Inc, 2001.*
3. *Charles B. Pfleeger, Shari Lawrence Pfleeger, "Security in Computing", Third Edition, Pearson Education, 2003.*

Course Outcome:

CO1	Provide security of the data over the network
CO2	Do research in the emerging areas of cryptography and network security
CO3	Implement various networking protocols.
CO4	Protect any network from the threats in the world.
CO5	Able to design firewall.